**GEBZE TECHNICAL UNIVERSITY**

**ELECTRONICS ENGINEERING DEPARTMENT**

ELM 366 COMMUNICATIONS LABORATORY

**REPORT 1**

# EXPERIMENT 1: DSB-LC and DSP-SC Amplitude Modulation

**Prepared by:** Esra Kirman

**Student Number:** 200102002028

**Team and Group Number:** Team 4

**Team Mates:** Fouad Aladhami, Büşra Ülker

**Date of the experiment: 14.03.2024**

# CONTENTS


1. **Introduction……3**
2. **Theoretical Background……3**
   - 2.1. **Amplitude Modulation……3**
   - 2.2. **Demodulation of AM Signals……..……6**
   - 2.3. **Double Side-band Modulation & Demodulation…….……..8**
3. **Experiment…….…….……...10**
   - 3.1. **Generating Amplitude Modulation (AM).…….……...10**
   - 3.2. **Demodulating an Amplitude Modulation (AM) Signal……..……18**
   - 3.3. **Generating a DSB Signal……22**
   - 3.4. **Demodulating an DSB Signal……25**
4. **Results……28**
5. **References…….……28**

## 1. Introduction

In this experiment, it is observed how Amplitude Modulation (AM), also known as Double Sideband Large Carrier (DSB-LC), is used in communication and how this signal can be demodulated. The modulation index is calculated for various modulation signals. The generation of Double Sideband Suppressed Carrier (DSB-SC) signal in communication and the understanding of the term "Suppressed Carrier" were examined. The demodulation of the transmitted message at the receiver and the effects of phase and frequency errors on the recovery process were investigated.

This experiment provides a basis for realizing AM modulation applications. Thus, the theoretical information seen can be observed in practical applications. Two-way communication radio systems, computer modems, citizen band radio, etc. The basics of the modulation technique used in the fields are discussed.

## 2. Theoretical Background

### 2.1. Amplitude Modulation

Modulation is the adaptation of the message signal to the channel. Amplitude modulation (AM) is one of the earliest modulation methods that is used in transmitting information over the radio. In amplitude modulation, transmission is achieved by increasing or decreasing the amplitude of the sent signal according to the signal and the desired requirements. This is changing the strength of the signal. In general, amplitude modulation definition is given as a type of modulation where the amplitude of the carrier wave varies in some proportion with respect to the modulating data or the signal. In AM, the carrier does not vary in amplitude. However, the modulating data is in the form of signal components consisting of frequencies either higher or lower than that of the carrier. [1]

There are four main types of amplitude modulation:

- Double Sideband-Suppressed Carrier Modulation (DSB-SC):
  $y(t) = m(t) * c(t) = A_m \cos(2\pi f_m t) * A_c \cos(2\pi f_c t)$
  $m(t)$: Message signal
  $A_m$: Amplitude of message signal
  $f_m$: Frequency of message signal
  $c(t)$: Carrier signal
  $A_c$: Amplitude of carrier signal
  $f_c$: Frequency of carrier signal
- Double Sideband- Large Carrier Modulation (DSB-LC):
  $y(t) = [1 + \mu m_n(t)] * c(t) = A_c * [1 + \mu m_n(t)] * \cos(2\pi f_c t)$
  $= A_c \cos(2\pi f_c t) + A_c \cos(2\pi f_c t)\, \mu m_n(t)$
  $m_n(t)$: $m(t)$ / maks$|m(t)|$ is called normalized message signal.
  $\mu$: Modulation index between 0 and 1.

- Single Sideband Modulation (SSB):
  $y(t) = m(t)\cos(2\pi f_c t) \mp m_h(t)\sin(2\pi f_c t)$
  $m_h(t)$: $m(t) * \frac{1}{\pi t}$ is called Hilbert Transform (where $*$ is convolution)
  In order to achieve single sideband modulation, the signal must be low around zero. The sound signal is a sign that provides this.
- Vestigial Sideband Modulation (VSB):
  Some signals, such as video, are not low around zero, unlike the audio signal. Therefore, vestigial sideband is used for video transmission. The DC value of audio signals is approximately zero, but video signals are high.

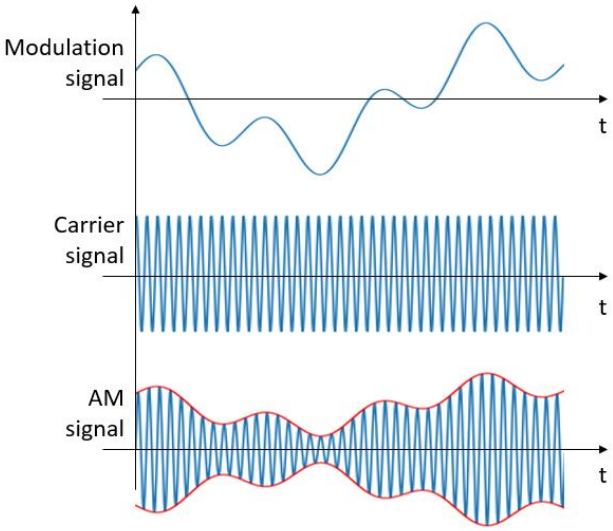



***Figure 1:*** *Amplitude Modulation [2].*

AM signal modulation is given in Figure 1. It can be seen that the carrier signal fluctuates between the modulated signal. The wave shown in red in the amplitude modulated signal (AM signal) is called the envelope. The envelope of a signal is a smooth curve outlining its extremes.

AM signal generation is given in Figure 2. First, a DC voltage value is added to the message signal. This represents message signals such as audio, video, etc. Then, modulation was made by multiplying it with the carrier signal and the modulated signal y(t) was obtained.

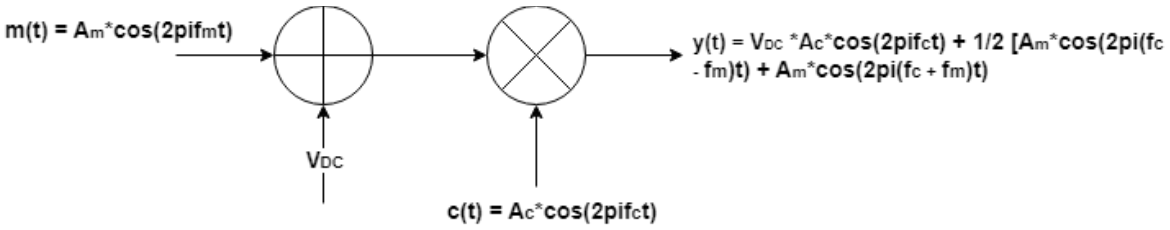



***Figure 2:*** *AM signal generation.*

As seen in the modulated y(t) signal, a dc component is formed at the carrier frequency $f_c$. This is the carrier itself. We see that this carrier is not visible in the suppressed carrier modulation technique. Because, as the name suggests, the carrier is suppressed (It is understood from the equation that when there is no DC component or it is very low, the carrier will have almost no component at the $f_c$ frequency.). In large carrier AM, the carrier appears at the $f_c$ frequency. Large carrier amplitude modulation can be applied to signals with a DC component.

The part of the envelope that restricts the carrier from the top is called upper envelope, and the part that restricts the carrier from the bottom is called lower envelope. The upper envelope is the message sign and the lower envelope is its inverted version. We can calculate the modulation index with the distances between these envelopes. Let's call the distance between the two envelopes of the point where the message signal gets its max value and the point where it gets its min value, A and B, respectively.

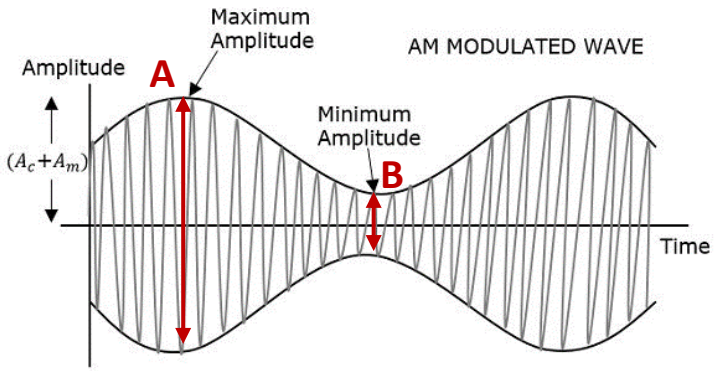



***Figure 3:*** *Modulated signal [3].*

Modulation index is the ratio of amplitude of the message signal to the amplitude of the plot tone with carrier frequency. In time domain modulation index $\mu$ can be computed as

$$\mu = \frac{A + B}{A - B}$$

It should not be forgotten that the modulation index is defined as a ratio of $\frac{A_m}{A_c}$. Where $C$ and $D$ are amplitudes of the frequency components belong to side band signals and the carrier signal, $\mu$ can be computed as $\mu = \frac{2C}{D}$ in frequency domain.

Reasons for modulation: high range transmission, quality of transmission and to avoid the overlapping of signals. For effective communication, the length of the antenna should be λ/4 times the modulating signal.

For instance audio signal's bandwidth is approximately 3kHz.

$$c = f.\lambda \rightarrow \lambda = \frac{c}{f} = \frac{3x10^8\ m/s}{3x10^3\ Hz} = 10^5 m$$

Antenna length: $l = \frac{\lambda}{4} = \frac{10^5 m}{4} = 25\ km$. When amplitude modulation is performed and the frequency is moved to 1 MHz, it will be seen that the minimum antenna length required will decrease to 75 m. Although it is not possible to produce a 25 km antenna, it is obvious that a 75 m antenna can be produced after modulation.

Advantages of amplitude modulation:

- Amplitude modulation is easier to implement.
- Demodulation can be done using a few components and a circuit.
- The receiver used for AM is very cheap. Even if the transmitter structure becomes more complex with large carriers, the receiver structure becomes simpler. Its simple receiver structure allows anyone to buy and use it. This is also suitable for commercial publications.

Disadvantages of amplitude modulation:

- It is not efficient about power usage.
- It requires a very high bandwidth.

### 2.2. Demodulation of AM Signals

After the signal is modulated, it is transferred to the receiver. When the modulated signal reaches the receiver, it is different from the original message signal. We need to obtain the message-carrying part of the modulated signal by passing it through some processes. Demodulation is the process by which the original information bearing signal is extracted from the overall received signal. There are different demodulations: demodulations with phase estimation (coherent demodulation) and demodulations without phase estimation (noncoherent demodulation). The demodulation scheme is given in Figure 4.

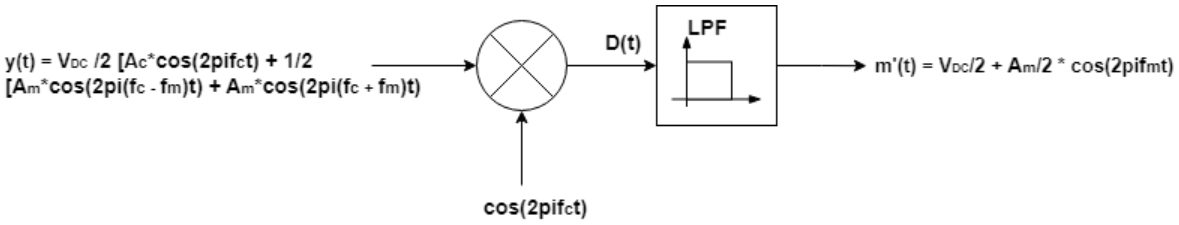



***Figure 4:*** *Demodulation.*

If the signal obtained as a result of modulation is multiplied by the carrier frequency at the receiver, the modulated signal is transferred to the base band. Then, when filtered with an appropriate low-pass filter, the message signal is obtained. Here the receiver needs to know the carrier frequency. The D(t) signal formed after the first process was obtained as shown below when the necessary mathematical operations were performed.

$$D(t) = \frac{V_{DC}}{2}\left(1 - cos(4pif_c t)\right) + \frac{A_m}{4} cos(2pi(2f_c - f_m)\, t) + \frac{A_m}{4} cos(2pi(2f_c + f_m)t) + 2cos(2pif_m t)$$

Covered here is the Envelope Detector, one of the simplest used by most portable radios etc. Let V(t) be the signal obtained when the y(t) signal is passed through a diode and RC circuit (Figure 5).

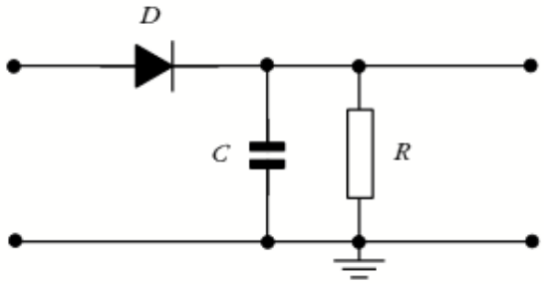



***Figure 5:*** *Envelope Detection [4].*

When the input wave's amplitude increases, the capacitor voltage is increased via the rectifying diode. When the input's amplitude falls, the capacitor voltage is reduced by being discharged by a 'bleed' resistor, R. The main advantage of this form of AM Demodulator is that it is very simple and cheap. Just one diode, one capacitor, and one resistor. That's why it is used so often [4].

B is bandwidth of message signal, $f_c$ is carrier frequency and $f_c \gg B$. Then, $RC$ must be between $\frac{1}{f_c} \ll RC \ll \frac{1}{B}$.

$$V(t) = \alpha\left[\frac{A_c}{\pi}\left(1 + \mu m_n(t)\right)\right] + Ripple = \alpha_1 + \alpha_2 m(t) + Ripple$$

We can add a capacitor to the output of the circuit to get rid of the DC frequency. We also pass it through a low pass filter to get rid of these high frequency components.

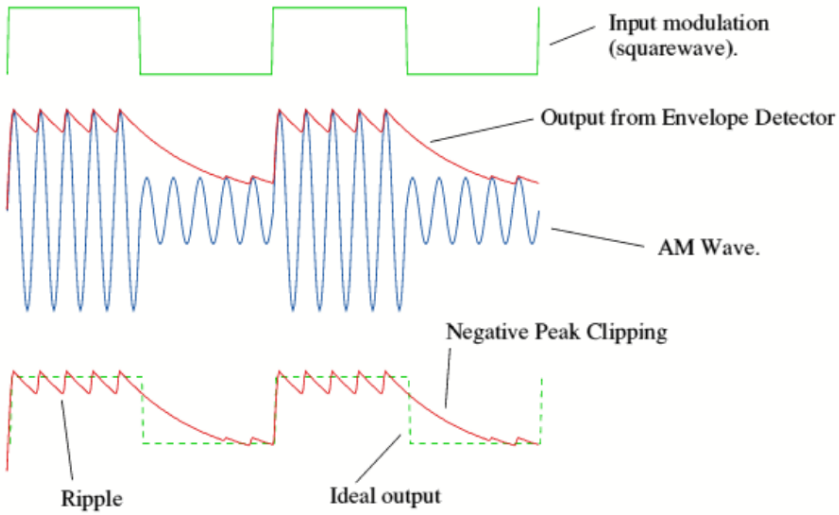



***Figure 6:*** *Ripple and Peak Clipping [4].*

### 2.3. Double Side-band Modulation & Demodulation

Double-sideband suppressed-carrier transmission (DSB-SC) is transmission in which frequencies produced by amplitude modulation (AM) are symmetrically spaced above and below the carrier frequency and the carrier level is reduced to the lowest practical level, ideally being completely suppressed. In the DSB-SC modulation, unlike in AM, the wave carrier is not transmitted; thus, much of the power is distributed between the side bands, which implies an increase of the cover in DSB-SC, compared to AM, for the same power use [5].

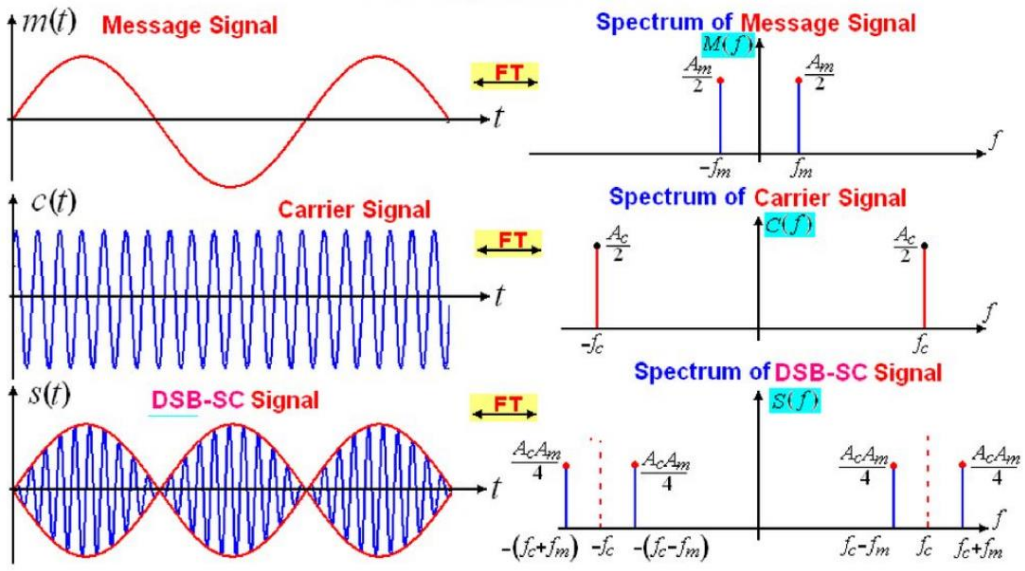



***Figure 7:*** *DSB-SC Modulation [6].*

DSB-SC signal can be produced with multiplying modulator, non-linear modulator and switching modulators. As seen in Figure 7, unlike amplitude modulation with large carriers, there is almost no component at the carrier frequency. The carrier is pressed.

DSB-SC Modulated signal: $y(t) = m(t) * c(t) = A_m \cos(2\pi f_m t) * A_c \cos(2\pi f_c t)$

$$y(t) = \frac{A_m A_c}{2}[\cos(2\pi(f_c - f_m t)) + \cos(2\pi(f_c + f_m t))]$$

DSBSC (Double Sideband Suppressed Carrier) efficiently utilizes the spectral bandwidth. Because the carrier signal is suppressed, the energy carried by the transmitted information is only spent on the sidebands. This allows for more information to be carried using less bandwidth. The disadvantage of DSBSC is that it requires a more complex circuit for demodulation on the receiver side. Therefore, DSBSC is generally used in radio communication systems, especially in systems with limited bandwidth.

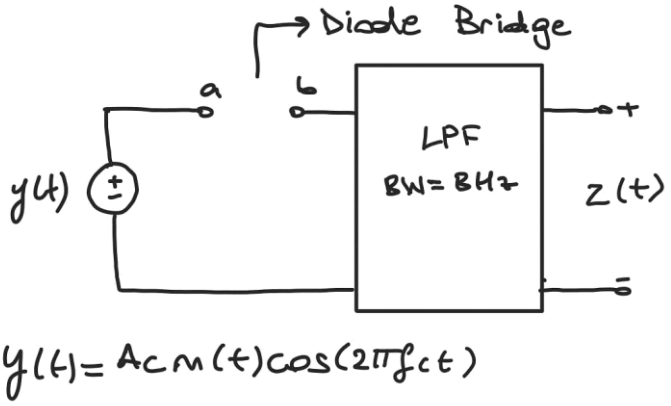



***Figure 8:*** *DSB-SC Demodulation.*

For DSBSC, Coherent Demodulation is done by multiplying the DSB-SC signal with the carrier signal (with the same phase as in the modulation process) just like the modulation process. This resultant signal is then passed through a low pass filter to produce a scaled version of the original message signal. However, demodulation of DSBSC requires a more complex circuit because the carrier signal is suppressed and it is important to take the correct operations to retrieve this suppressed carrier signal. First, let's do the multiplication (D(t)) and then pass it through LPF.

$$D(t) = y(t)\cos(2\pi f_c t) = A_c A_m/2 \cos(2pif_m t) + A_c A_m/2 \cos(2pif_m t)\cos(4\pi f_c t)$$

$$D(t) = \frac{A_c A_m}{2}\cos(2pif_m t) + \frac{A_c A_m}{4}[\cos(2pi(2f_c - f_m)t) + \cos(2pi(2f_c + f_m)t)]$$

After LPF $\rightarrow m'(t) = \frac{A_c A_m}{2}\cos(2pif_m t)$

## 3. Experiment

### 3.1. Generating Amplitude Modulation (AM)

The connections made are given in Figure 9. The following adjustments were made via the Tetra BB control interface:

Frequency (tone) = 2 kHz Variable DC = -4 V **(Since the DC voltages on the board we used are connected in reverse, +4V was connected to give -4V),** Gain (g) = 0.25, Gain (G) = 0.25 (gains of the adder block), Frequency (carrier) = 100 kHz. In the oscilloscope panel: Channel 0 Voltage Range: 1V/div Channel 1 Voltage Range: 500mV/div Horizontal Time base: 100µs/div Trigger: Edge, Channel 0 Source **(On the trigger screen, it was written that channel 1 should be set, but it could not work because channel 1 was below the trigger. That's why channel 0 was chosen as the trigger and the signal started to shown in screen),** Level:1,2V Horizontal Position(%):10 Probe Attenuation :1x

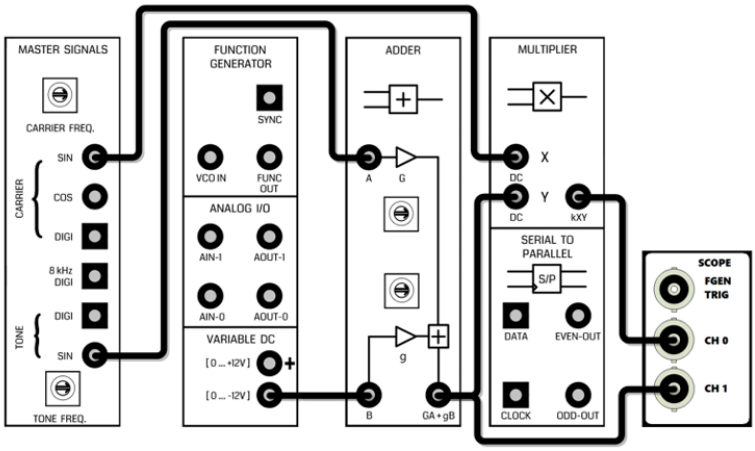



***Figure 9:*** *Generate an AM signal*

As can be seen from the connections, the sinusoidal message signal is summed with -4V DC value and the message sign is created. This signal is then multiplied by the carrier and modulated. The modulated signal is read with channel 0. In Channel 1, the DC added message signal is examined. As can be seen in the output graph in Figure 10, the message sign has taken the form of an envelope, as explained in the theoretical part. In the following figures, max and min ranges were read with a cursor and the modulation index was calculated accordingly.

$$m(t) = V_{DC} + A_m \, sin(2\pi f_m t)$$

$$y(t) = [V_{DC} + A_m \, sin(2\pi f_m t)] * \sin(2\pi f_c t)$$

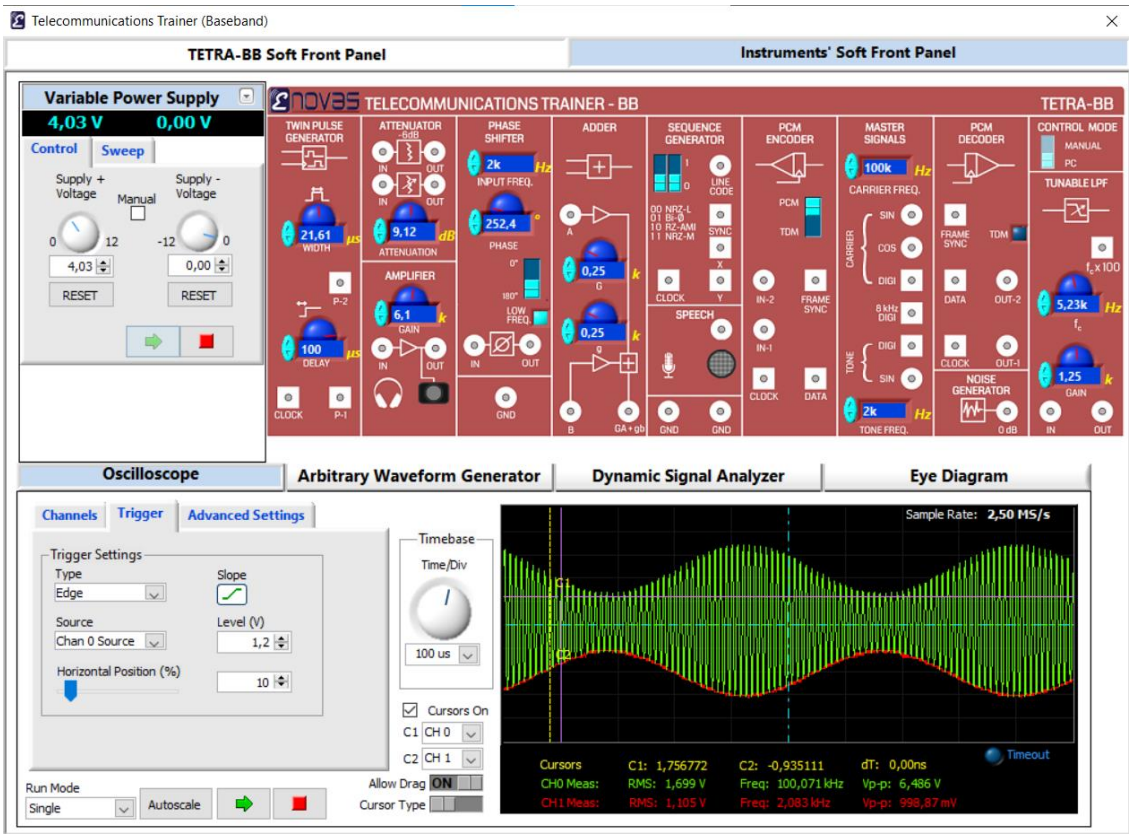



***Figure 10:*** *AM signal*

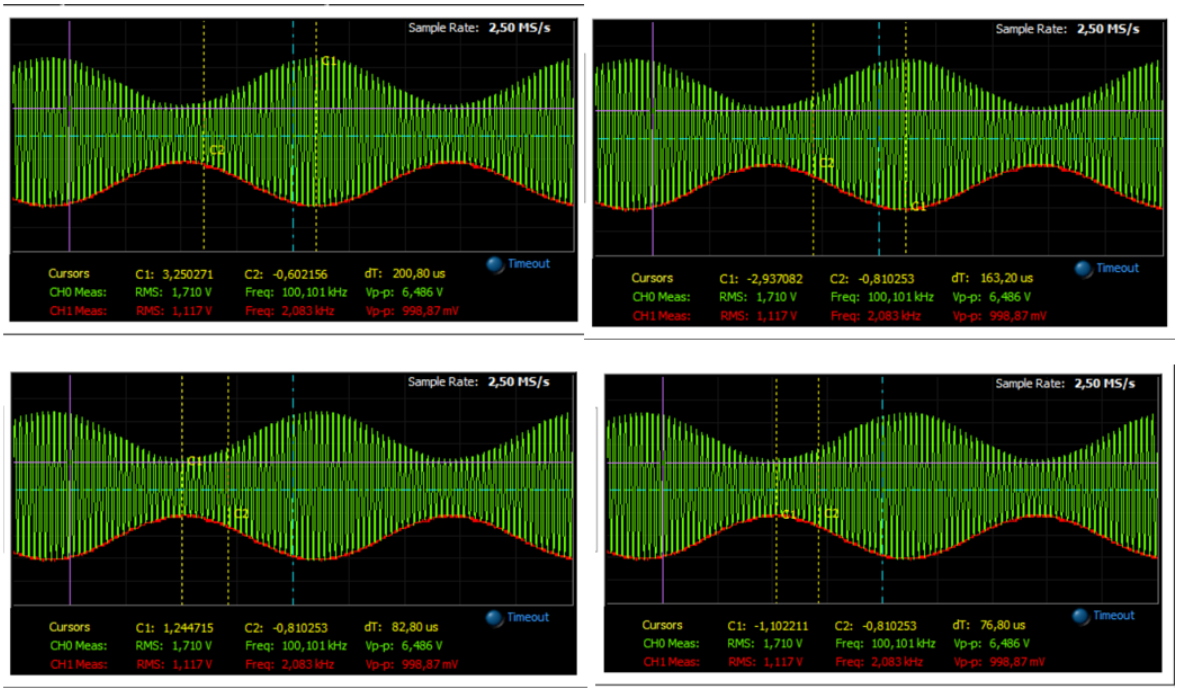



***Figure 11:*** *Max. and min. value of AM signal.*

As can be seen in Figure 11: $A_{max}$= 3.25, $A_{min}$= -2.93, $B_{max}$= 1.24, $B_{min}$= -1.10.

$$\mu = m = \frac{A-B}{A+B} = \frac{\big(3.25-(-2.93)\big)-\big(1.24-(-1.10)\big)}{\big(3.25-(-2.93)\big)+\big(1.24-(-1.10)\big)} = \frac{6.18-2.34}{6.18+2.34} = 0.45$$

The amplitude of the message (Am) was increased to a level where the modulation index was equal to 1 (G gain). However, due to circuit factors, we could not reach exactly 1 as expected. An approximate value was obtained with G=0.5. This adjustment is given in Figure 12.

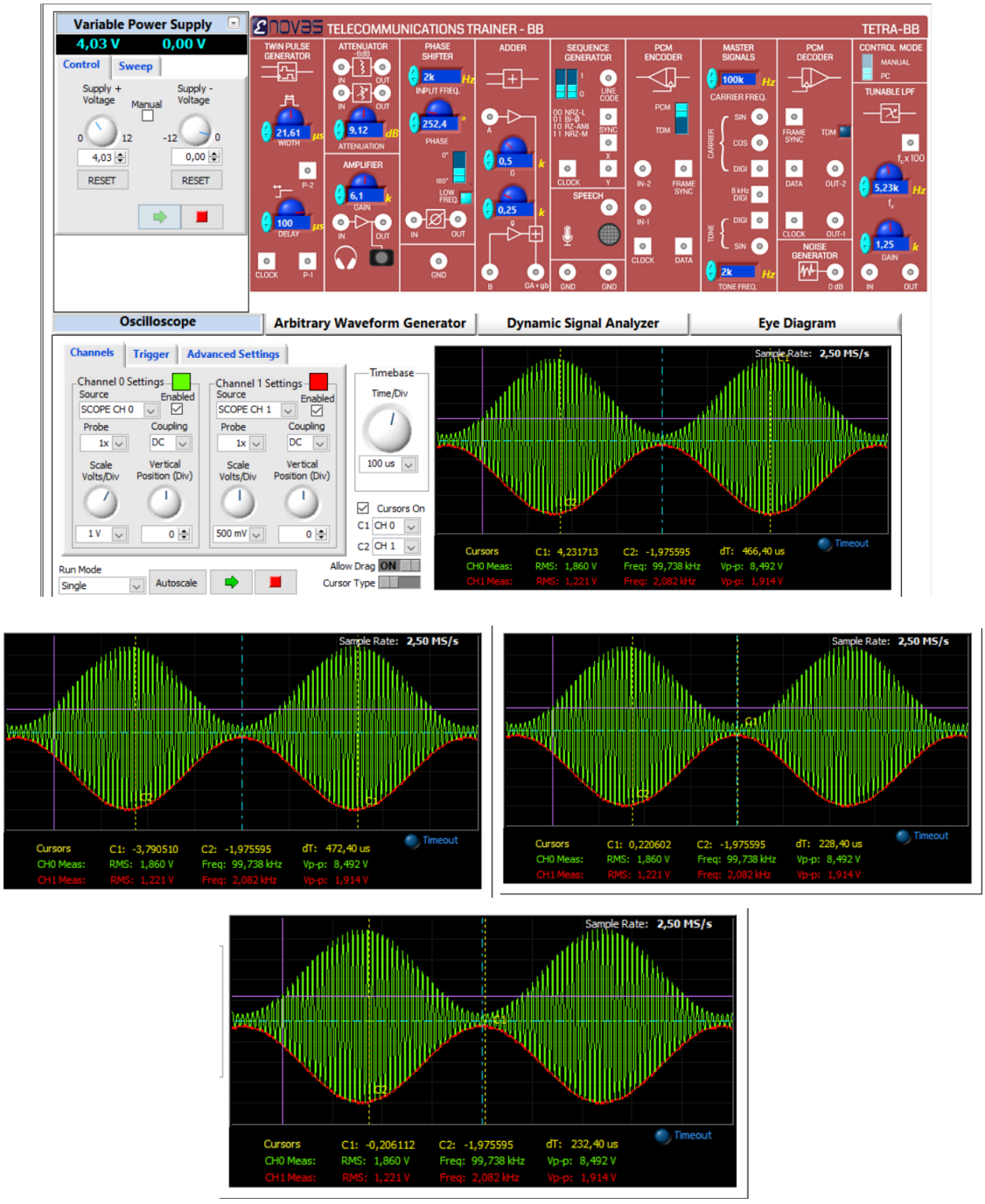



***Figure 12:*** *Max. and min. value of AM signal for m=1.*

As can be seen in Figure 12: $A_{max}$= 4.23, $A_{min}$= -3.79, $B_{max}$= 0.22, $B_{min}$= -0.20.

$$\mu = m = \frac{A-B}{A+B} = \frac{\big(4.23-(-3.79)\big)-\big(0.22-(-0.20)\big)}{\big(4.23-(-3.79)\big)+\big(0.22-(-0.20)\big)} = \frac{8.02-0.42}{8.02+0.42} = 0.90$$

The amplitude of the message (Am) was increased to a level where the modulation index was equal to 1.5 (G gain). However, due to circuit factors, we could not reach exactly 1.5 as expected. An approximate value was obtained with G=0.75. This adjustment is given in Figure 13.

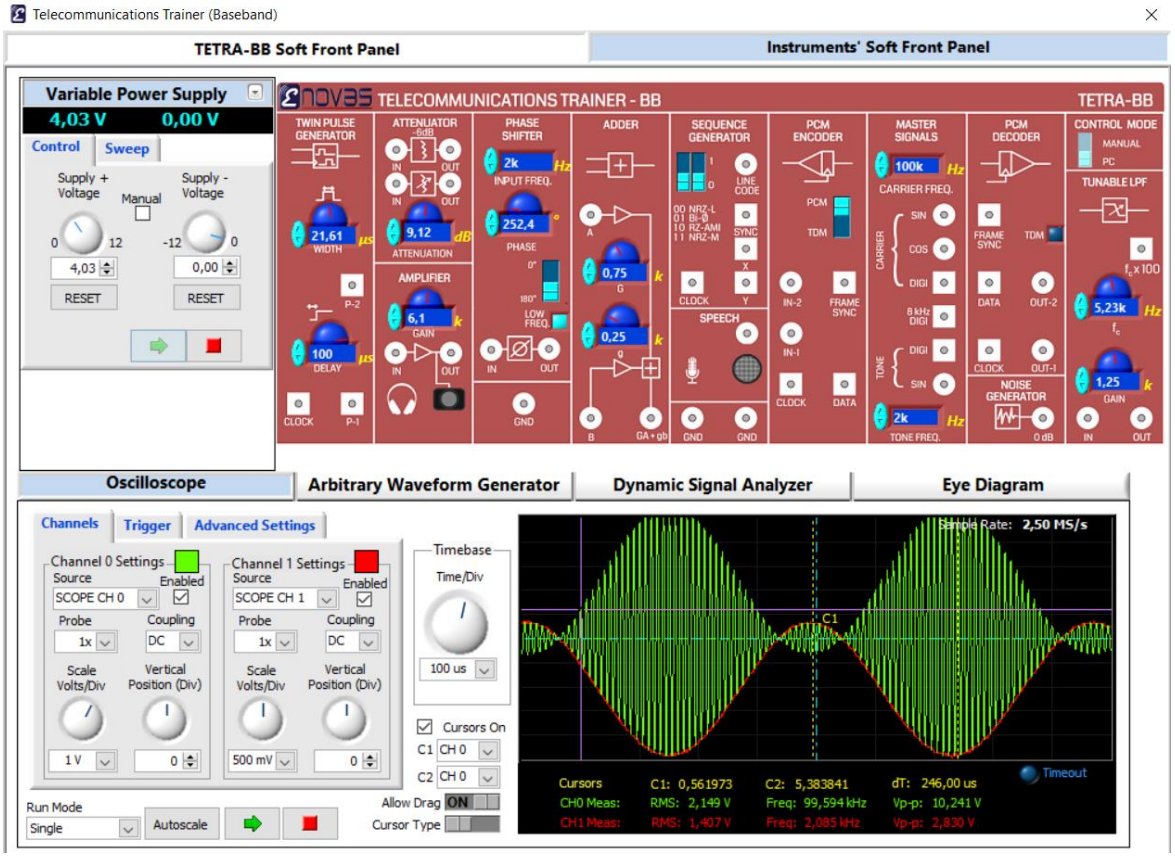



***Figure 13:*** *Max. and min. value of AM signal for m=1.5.*

As can be seen in Figure 13: A=5.38*2 = 10.76, B = 0.56*2 = 1.12

$$\mu = m = \frac{A-B}{A+B} = \frac{10.76-1.12}{10.76+1.12} = 1.25$$

As the amplitude of the message signal is increased, the envelope begins to exceed its limits. This is called overmodulation. This causes distortion in the signal. This distortion can make it difficult to demodulate the original message signal correctly. For this reason, the modulation index is generally kept between 0 and 1 (as mentioned in theoretical part).

The gains were adjusted so that the modulation index was 0.5 and the message frequency was increased to 5kHz:

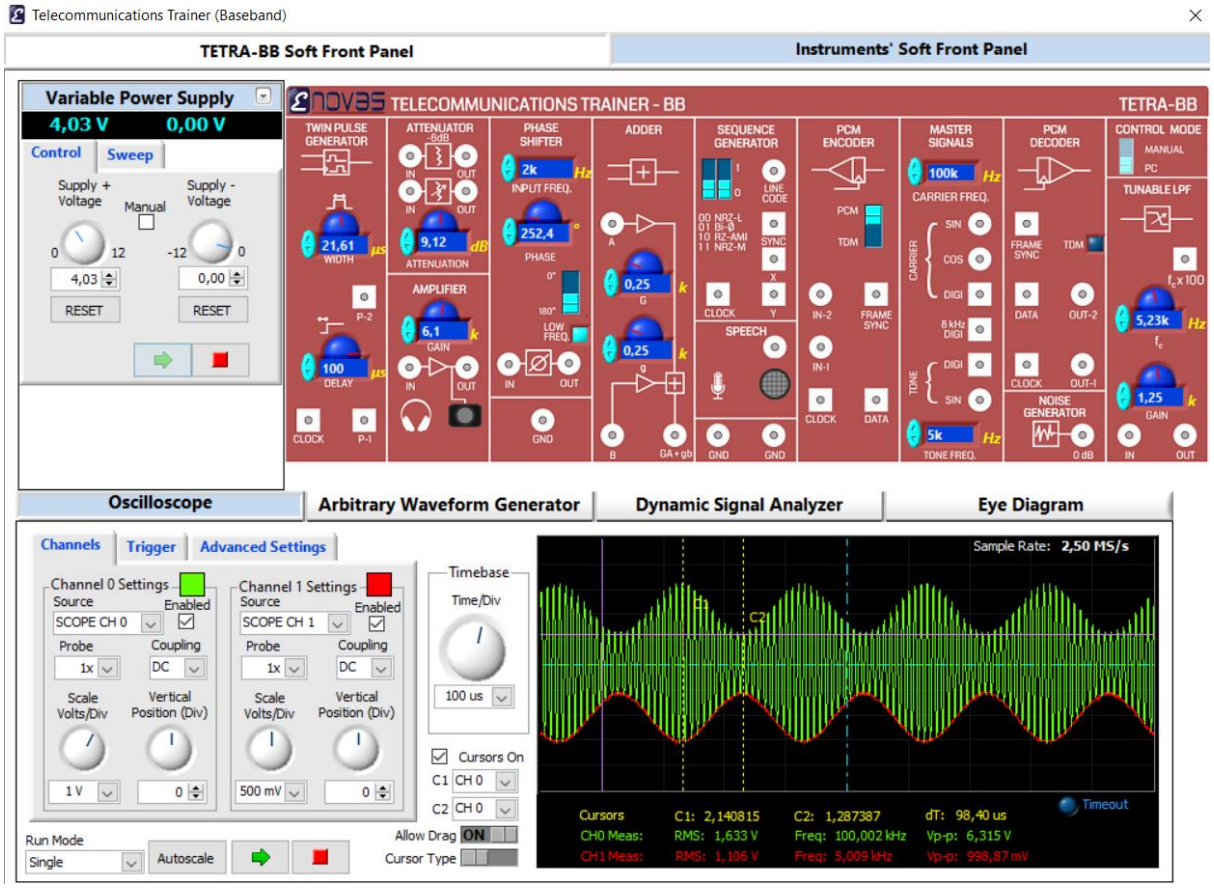



***Figure 14:*** *AM signal for m=0.5 and* $f_m = 5kHz$.

As seen in Figure 14, A and B values did not change, the modulation index was 0.45, a close result was found as 0.41 (slight differences are due to the differences in placing the cursor). Because in amplitude modulation, the amplitude modulation index depends only on the amplitudes. The change in frequency affects the index in frequency modulation.

As can be seen in Figure 13: A=3.04 *2 = 6.08, B = 1.28*2 = 2.56

$$\mu = m = \frac{A - B}{A + B} = \frac{6.08 - 2.56}{6.08 + 2.56} = 0.41$$

The dynamic signal analyzer tab was opened and the following settings were made: Input Settings: Channel 0 Center Frequency: 100000 Frequency Span: 60000 Mode: RMS Resolution (lines): 1600 Weighting: Exponential Window: 7 Term-B Harris # of Averages: 5

As expected in large carrier amplitude modulation, a dirac delta signal appears at the carrier frequency $f_c = 100kHz$ , and dirac outputs of the message signal appear at $f_c$-$f_m$ = $95kHZ$ and $f_c$-$f_m$ = 105 $kHz$. As can be read from the graph, the amplitudes are -10 for the message signal and +2 for the carrier.




***Figure 15:*** *AM signal's spectrum for* $f_c = 100kHz$ *and* $f_m = 5kHz$.

When $f_m = 2kHz$, the carrier frequency was increased from 80kHz to 120 kHz in 20kHz steps.

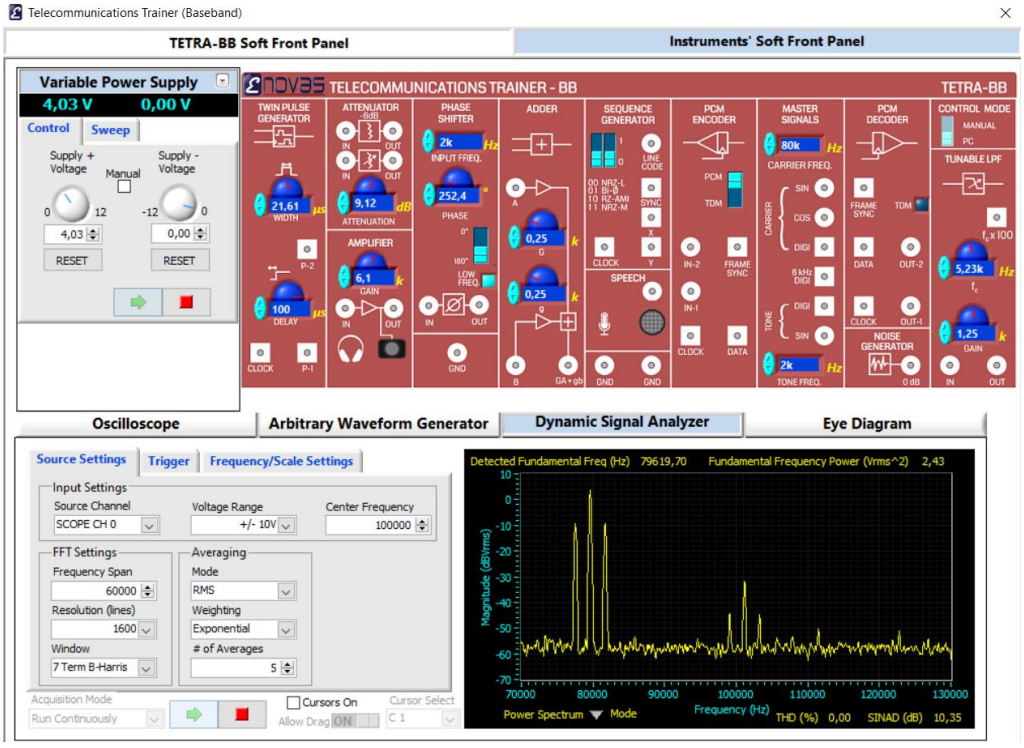



***Figure 16:*** *AM signal's spectrum for* $f_c = 80kHz$ *and* $f_m = 2kHz$

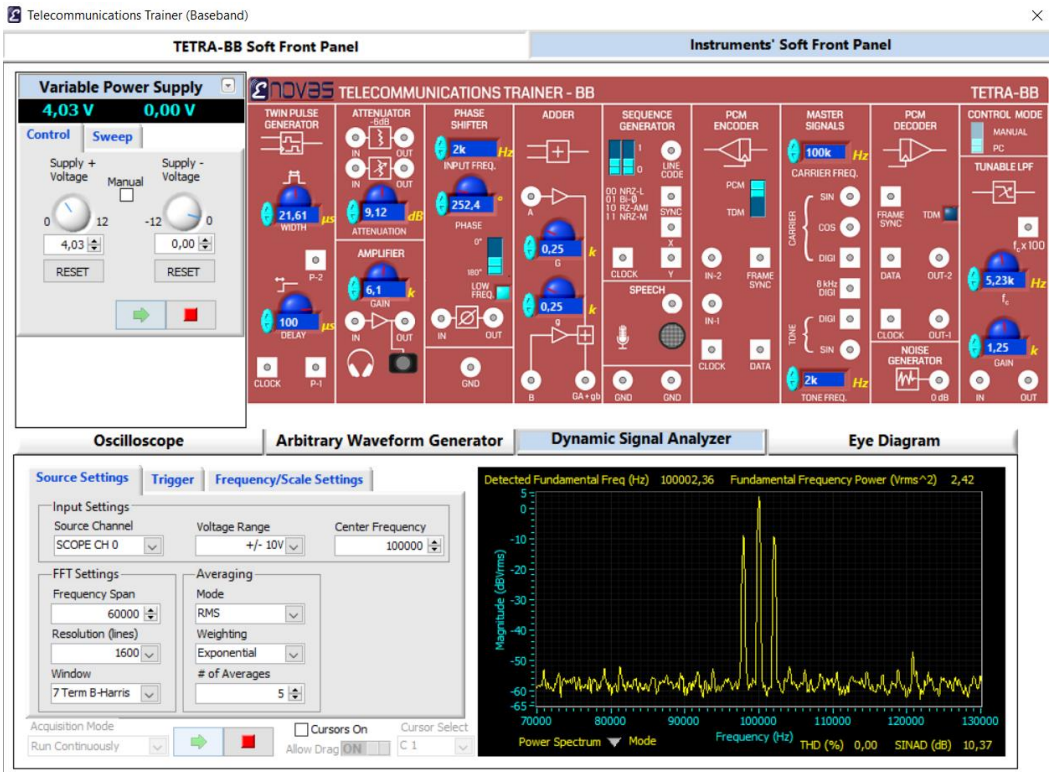



***Figure 17:*** *AM signal's spectrum for* $f_c = 100kHz$ *and* $f_m = 2kHz$




***Figure 18:*** *AM signal's spectrum for* $f_c = 120kHz$ *and* $f_m = 2kHz$

Figures 16,17,18 show the change that occurs when the carrier frequency increases while the message signal frequency is constant. In the frequency domain, the signal reaches the frequency band where the carrier frequency is located.

When $f_c = 100kHz$, the message frequency was increased from 1kHz to 5kHz in 2kHz steps.

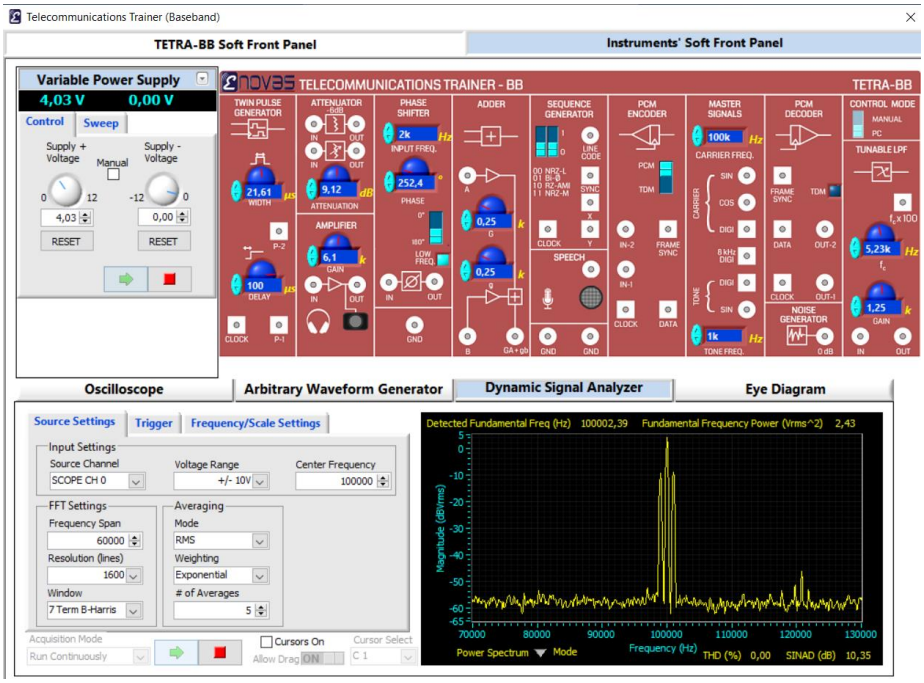



***Figure 19:*** *AM signal's spectrum for* $f_c = 100kHz$ *and* $f_m = 1kHz$

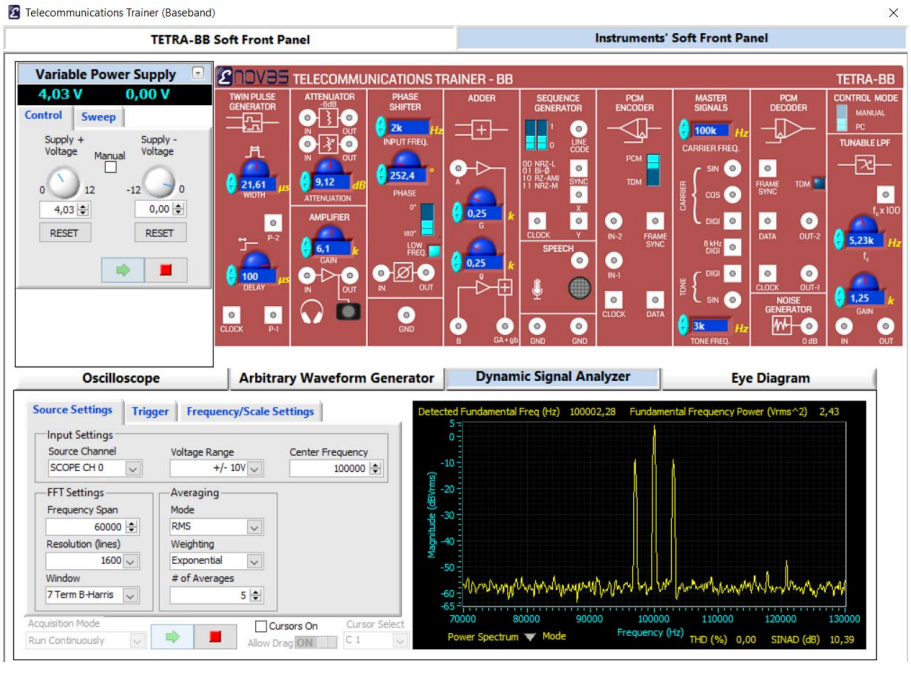



***Figure 20:*** *AM signal's spectrum for* $f_c = 100kHz$ *and* $f_m = 3kHz$

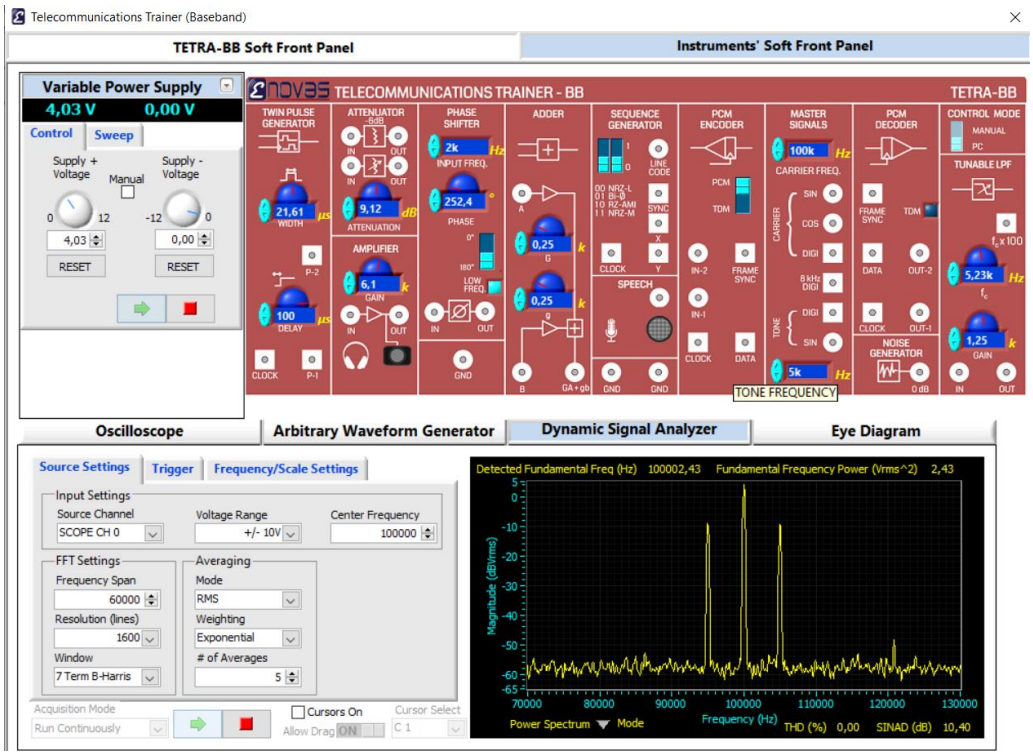



***Figure 21:*** *AM signal's spectrum for* $f_c = 100kHz$ *and* $f_m = 5kHz$

Figures 19,20,21 show the change that occurs when the message frequency increases while the carrier signal frequency is constant. As the difference between $f_c$-$f_m$ and $f_c$+$f_m$ changes in the frequency domain, we see that the dirac outputs diverge as $f_m$ increases.

### 3.2. Demodulating an Amplitude Modulation (AM) Signal

The oscilloscope panel was opened and the following settings were made: Channel 0 Voltage Range: 1V/div Channel 1 Voltage 200mV/div Range: Horizontal Time base: 100µs/div Trigger: Edge, Chan 1 Source, Level:1,2V Horizontal Position (%): 10. In some cases, scales were changed to make the output easier to see.

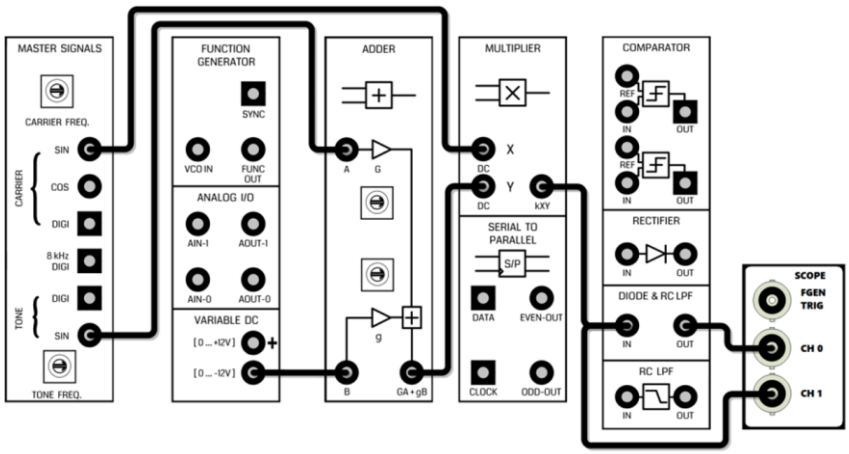



***Figure 22:*** *Connection of envelop detector demodulation.*

When the connections in Figure 22 are examined, it is seen that demodulation is made with the envelope detector structure. The generated signal is inserted into the DIODE & RC LPF and the demodulated output is read with channel 0.

In the figures below, the cases where the modulation index is 0.5, 1 and 1.5 are examined.

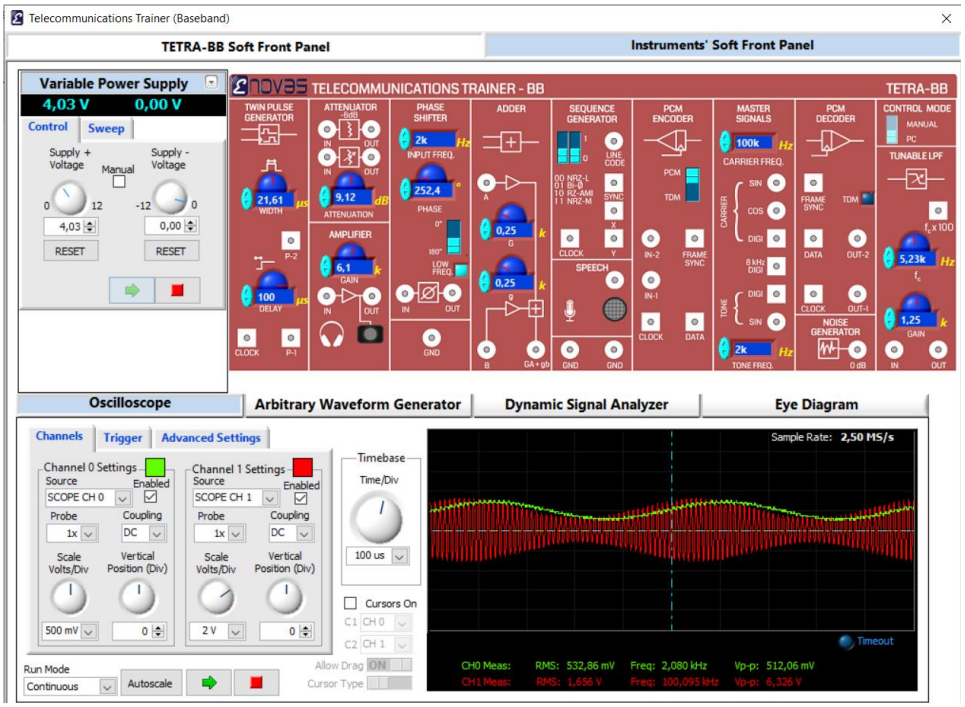



***Figure 23:*** *Demodulation when modulation index m=0.5.*

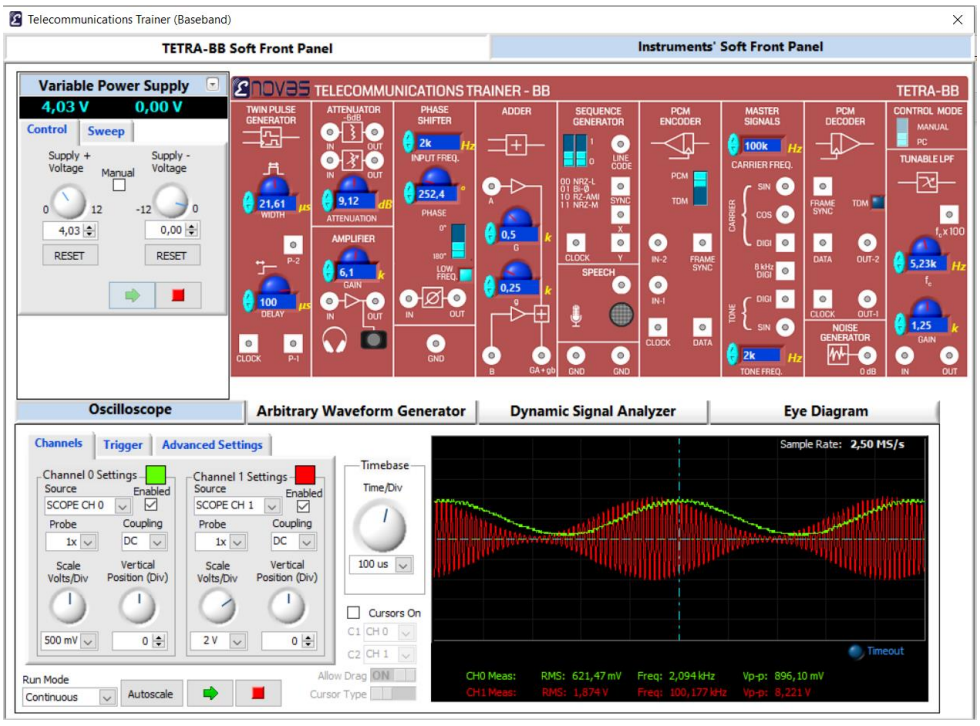



***Figure 24:*** *Demodulation when modulation index m=1.*

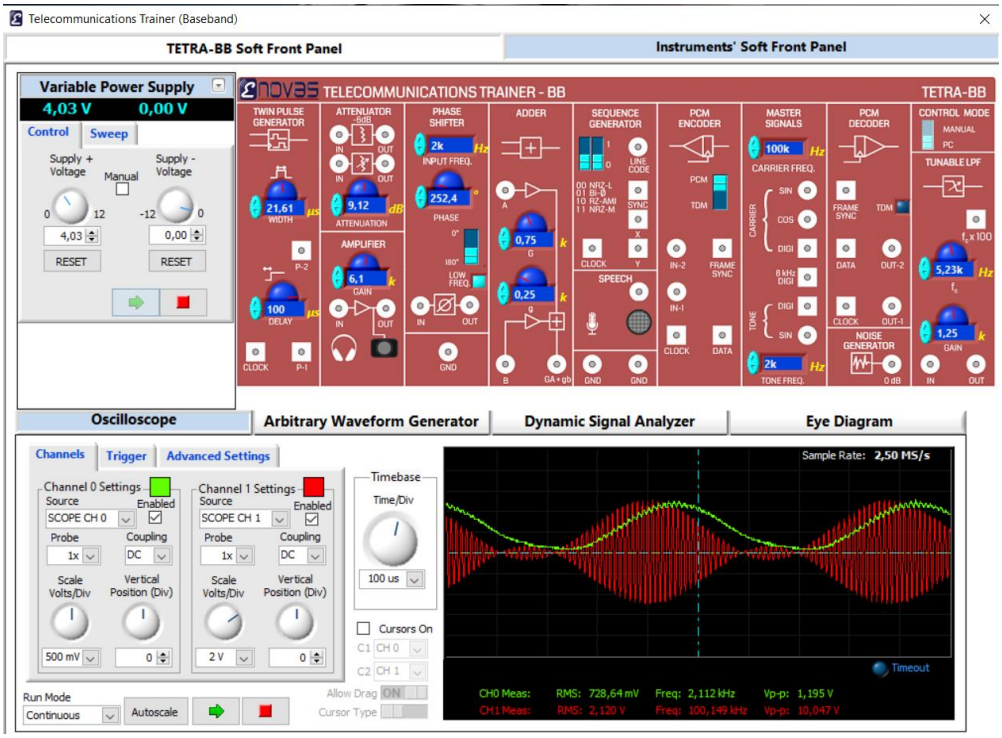



***Figure 25:*** *Demodulation when modulation index m=1.5.*

The modulated signal in channel 1 is measured in its demodulated form in channel 0. As can be seen, when the modulation index exceeds 1, the signal begins to deteriorate. Distortion usually begins when the modulation index exceeds 1. Because at this point, the modulated signal rises above the amplitude of the carrier signal and the signal is crushed or cut off. For example, when the modulation index is 1, the amplitude of the message signal (Am) becomes equal to the amplitude of the carrier signal (Ac).

Demodulation of an AM signal can also be done with a mixer by using the same carrier signal.

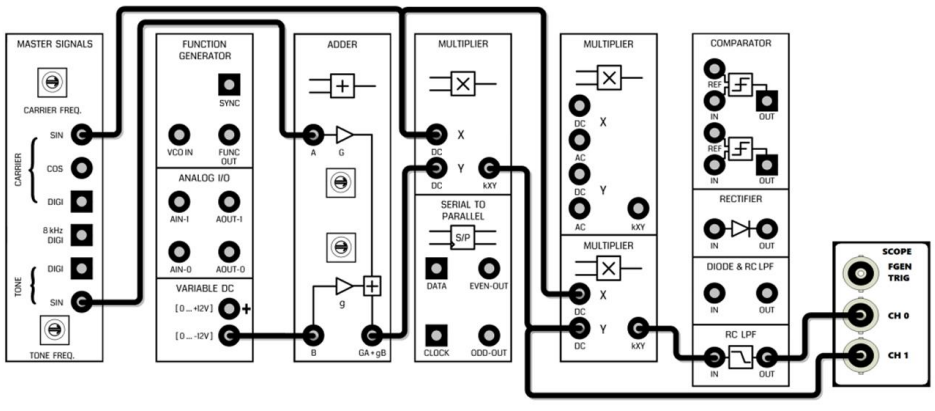



***Figure 26:*** *Demodulation with RC LPF.*

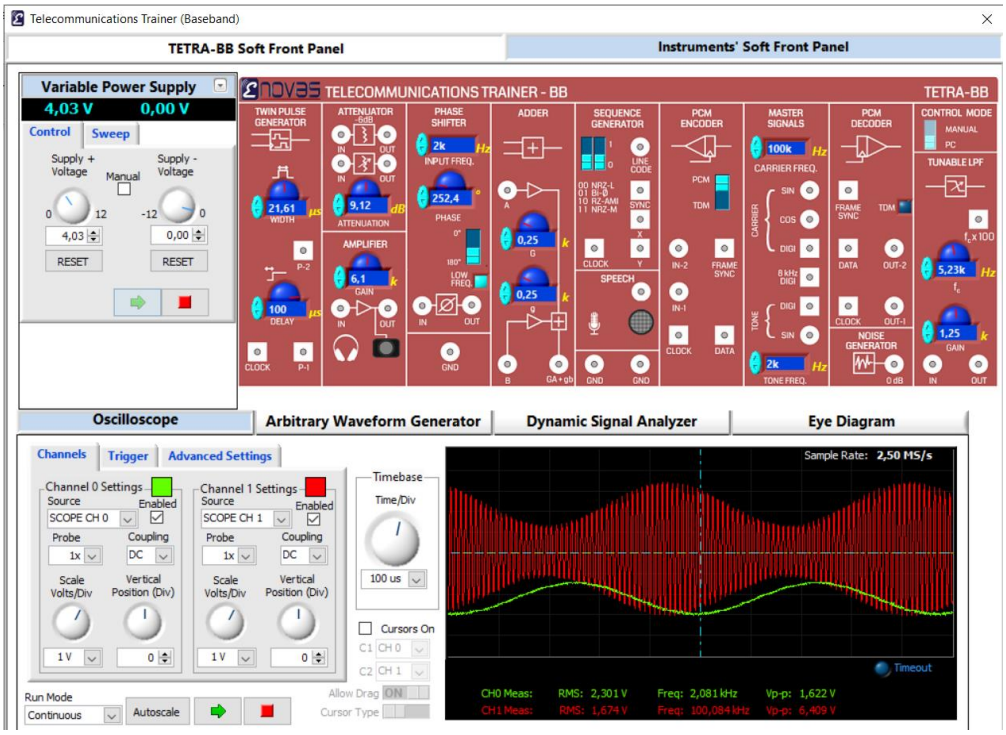



***Figure 27:*** *RC LPF Demodulation when modulation index m=0.5.*

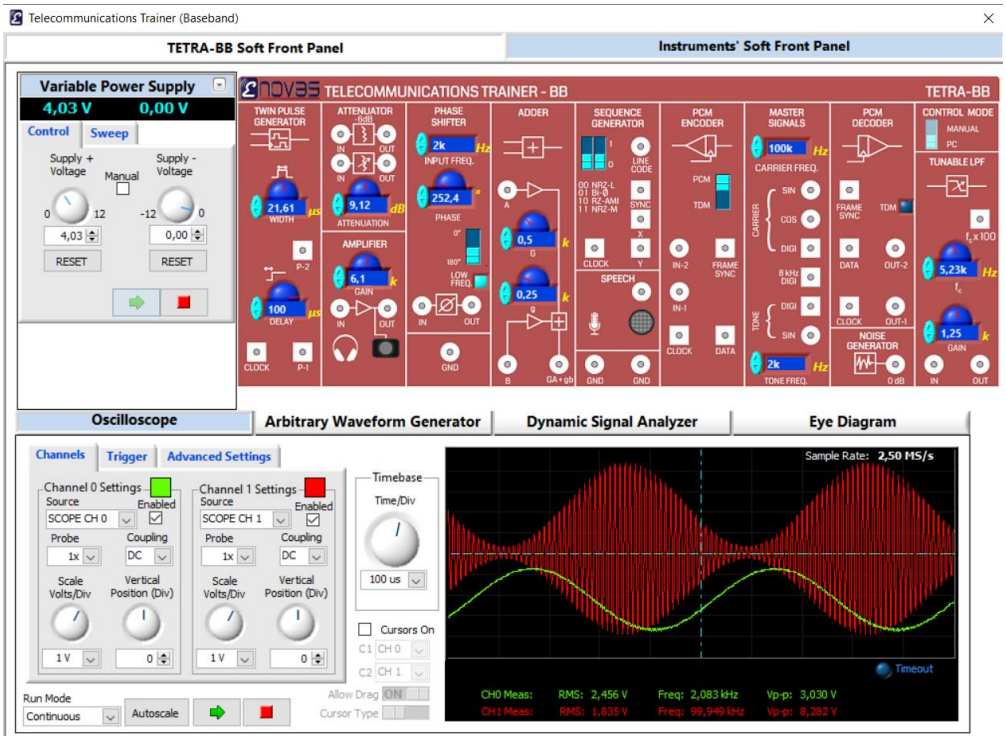



***Figure 28:*** *RC LPF Demodulation when modulation index m=1.*

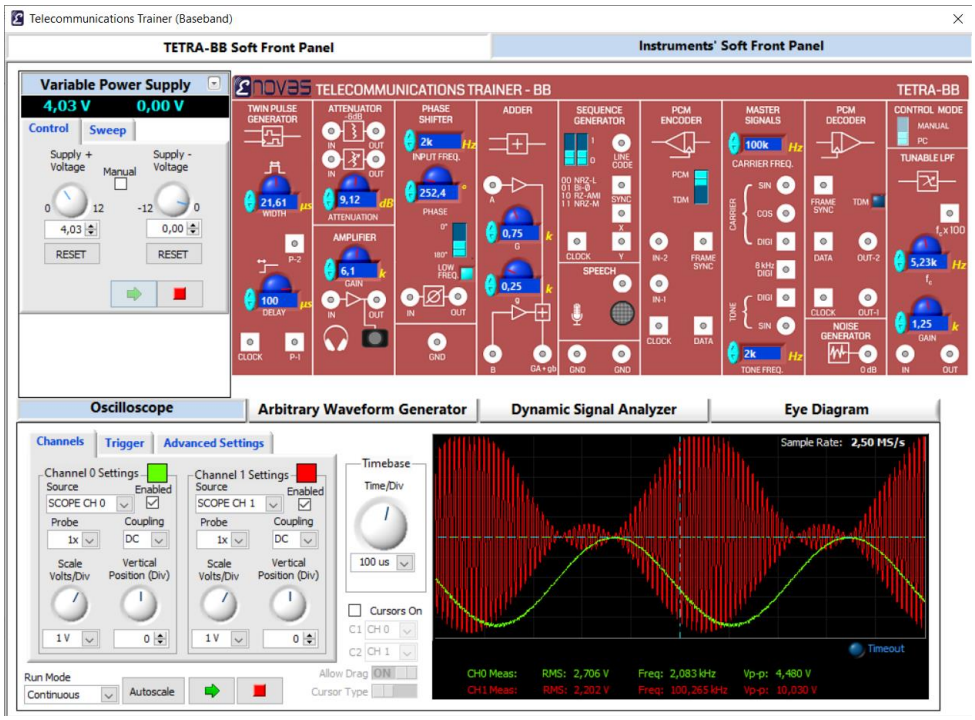



***Figure 29:*** *RC LPF Demodulation when modulation index m=1.5.*

### 3.3. Generating a DSB Signal

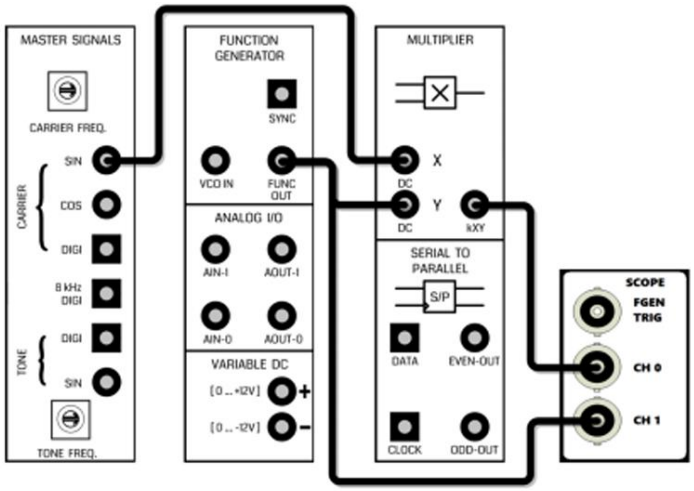



***Figure 30:*** *DSB-SC Modulation*

As seen in Figure 30, the sinusoidal message wave is modulated by multiplying it with the carrier. This modulation is suppressed carrier modulation.

$$y(t) = m(t) * c(t) = A_m \, sin(2\pi f_m t) * sin(2\pi f_c t)$$

The following settings were made: Peak to peak amplitude of the carrier is Vpp = 4 V, the carrier frequency (fc) to 100 kHz. Message frequency fm to 2 kHz and peak to peak amplitude to Vpp = 2 V of the message signal m(t) by using the Function Generator block with the following settings: Frequency: 2kHz Amplitude: 2 Vpp Modulation type: None DC Offset:0 V. The oscilloscope panel: Channel 0 Voltage Range: 1V/div Channel 1 Voltage Range: 500mV/div Horizontal Time base: 100µs/div Trigger: Immediate

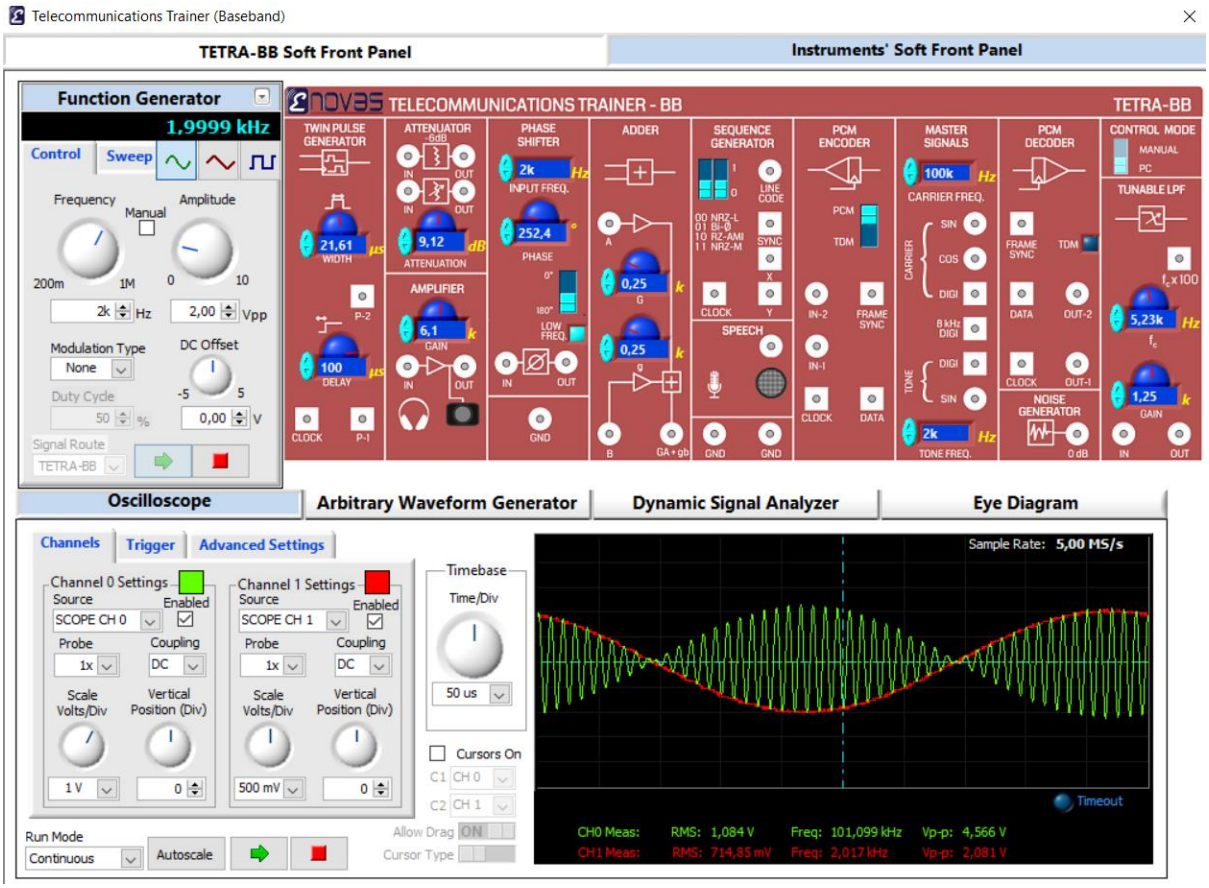



***Figure 31:*** *DSB-SC Modulated signal.*

There is no DC component in the message signal. In SC AM, the modulated signal is produced without a carrier as the amplitude of the carrier signal is zero. That is, the modulation process is done only with the amplitude of the message signal.

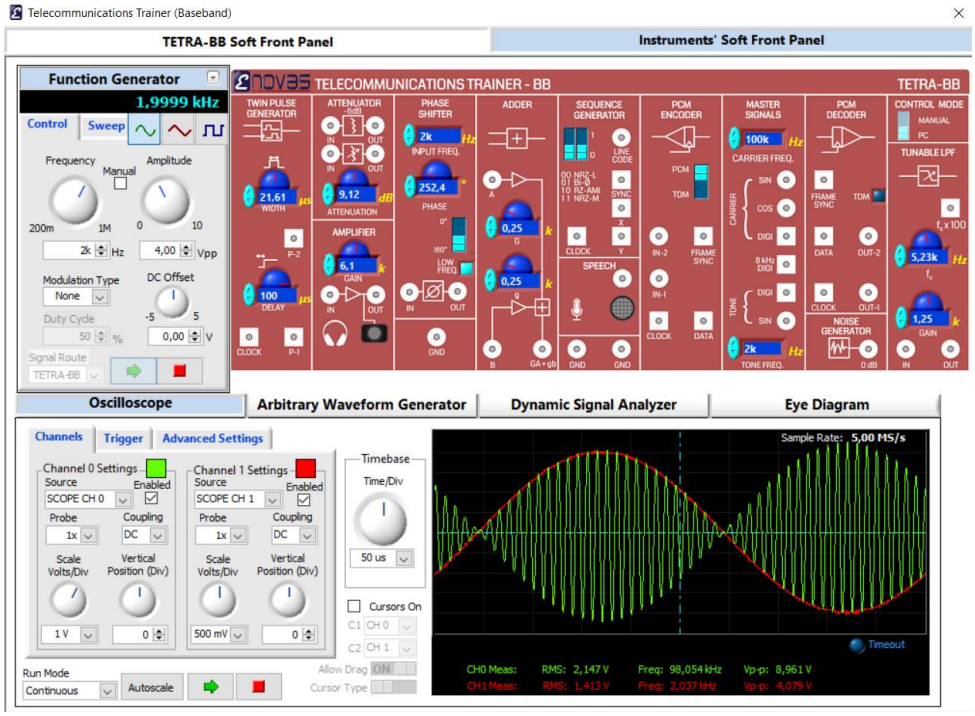



***Figure 32:*** *DSB-SC Modulated signal when Vpp = 4V*

When the Vpp value increased, the amplitude of the modulated signal increased to 2 times Vpp. The amplitude of the message signal directly affects the output. However, since the carrier is suppressed, it does not affect the amplitude of the carrier.

The DSA panel was opened and the following settings were made: Input Settings: Channel 0 Center Frequency: 100000 Frequency Span: 20000 Mode: RMS Resolution (lines): 3200 Weighting: Exponential Window: 7 Term-B Harris # of Averages: 5

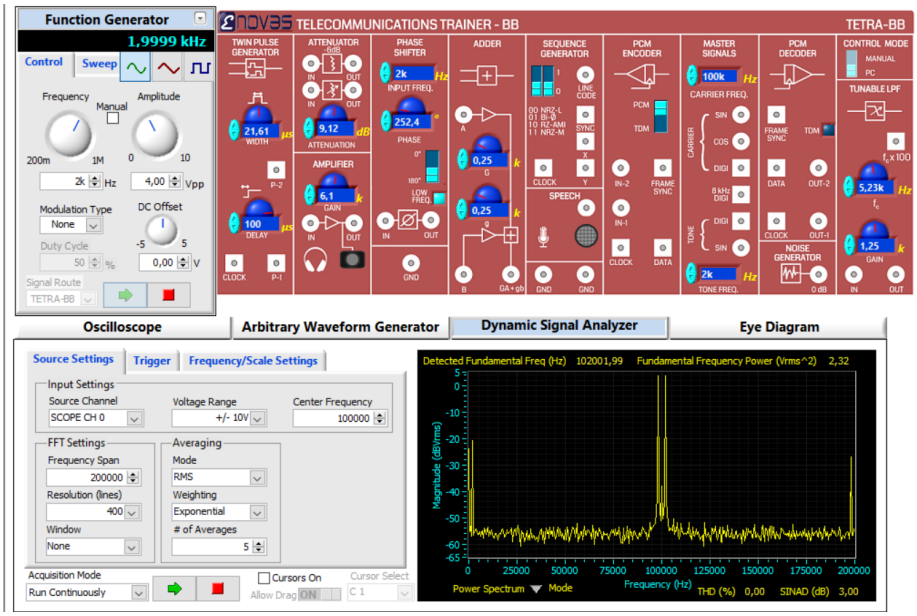



***Figure 32:*** *DSB-SC Modulated signal spectrum.*

The scale has been changed to see the spectrum more easily.

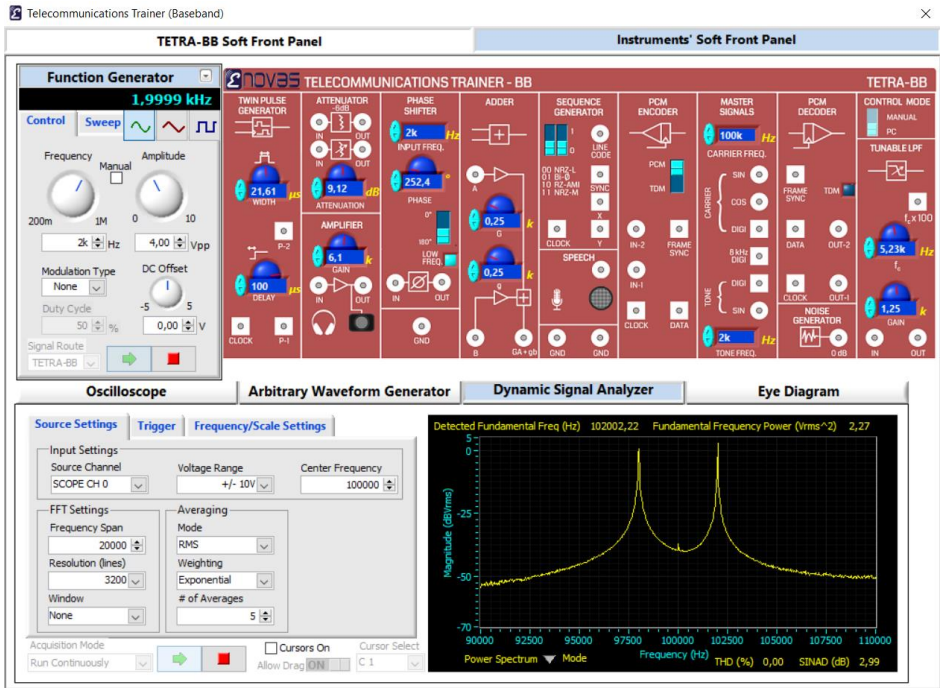



***Figure 33:*** *DSB-SC Modulated signal spectrum scaled.*

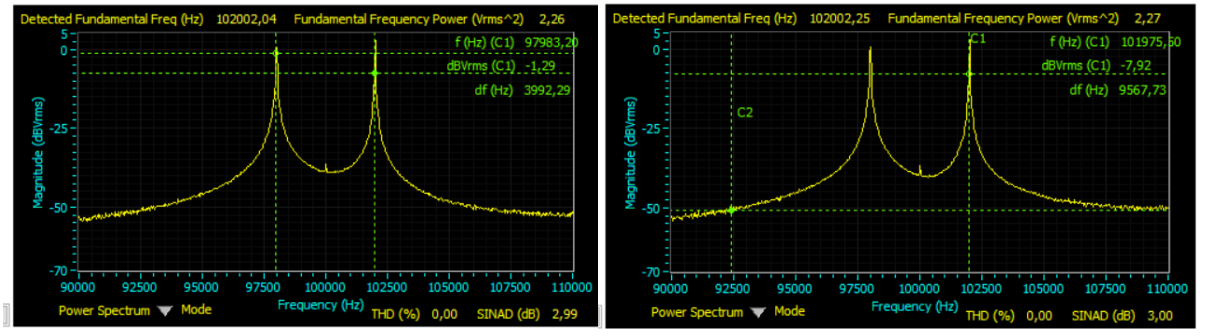



***Figure 34:*** *DSB-SC Modulated signal spectrum frequency.*

As can be seen from the results, while there is no impulse at the carrier frequency, impulses of the message signal are visible. In suppressed carrier AM, only the message signal is present. The carrier is pressed onto its name (suppressed carrier).

### 3.4. Demodulating an DSB Signal

RC LPF was used for DSB-SC modulation as shown in Figure 35. The demodulated signal is given in figure 36. CH0 indicates the demodulated signal. CH1 is the message signal.

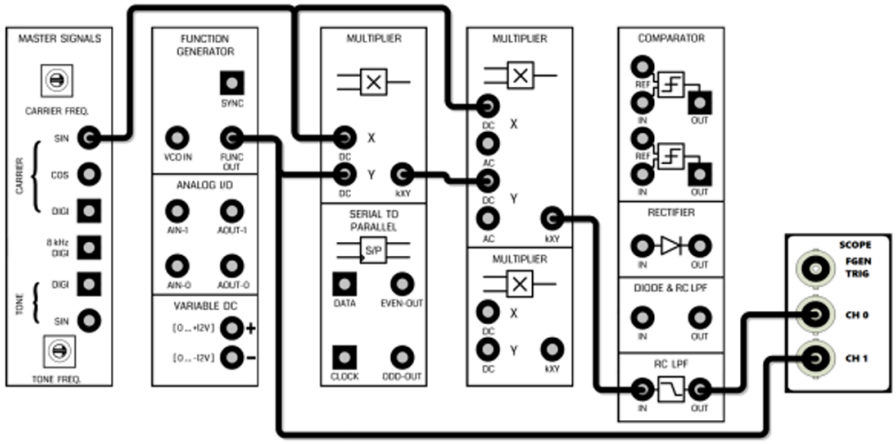



***Figure 35:*** *DSB-SC Demodulation.*

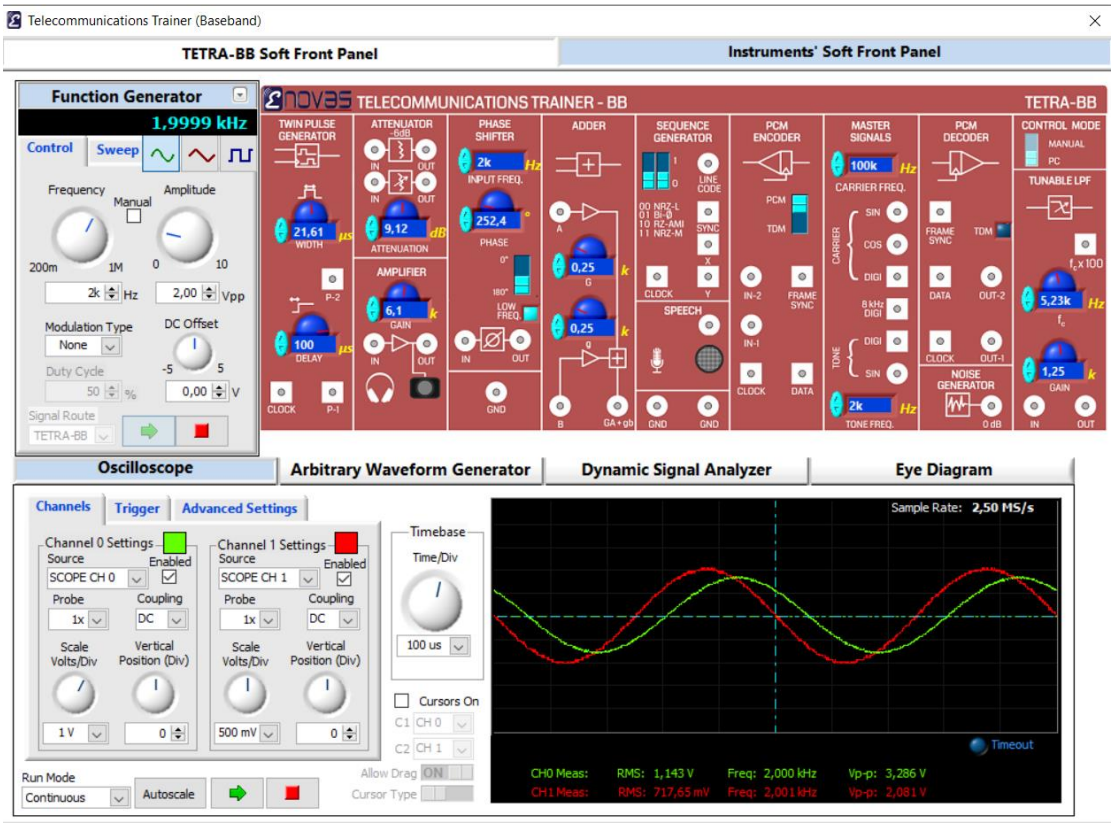



***Figure 36:*** *DSB-SC Demodulated signal.*

The oscilloscope panel was opened and the following settings were made: Channel 0 Voltage Range: 1V/div Channel 1 Voltage Range: 500mV/div Horizontal Time base: 100µs/div Trigger: Immediate. Carrier frequency (fc) to 100 kHz. Message frequency fm to 2 kHz and peak to peak amplitude to Vpp = 2 V of the message signal m(t).

The frequency of modulated and demodulated signals is shown in figure 36. Both are sinusoidal waves with a frequency of 2kHz. The demodulated signal carries the message.

The DSA panel was opened and the following settings were made: Input Settings: Channel 0 Center Frequency: 5000 Frequency Span: 10000, Mode: RMS Resolution (lines): 1600 Weighting: Exponential Window: 7 Term-B Harris # of Averages: 5




***Figure 37:*** *DSB-SC Demodulated signal spectrum.*

In Figure 37, we see that there is an impulse at 2 kHz. In demodulation: Since the DSB-SC signal is negative in the negative half-period, the diode becomes conductive only in the negative half-periods. The fact that the diode only conducts in the negative half-periods gives us the two side bands created by the DSB-SC signal in which the carrier signal is suppressed. The capacitor amplifies the amplitude of this signal. We fix it and obtain the modulation signal.

## 4. Results

Overmodulation in the signals was observed. When the modulation index exceeds 1 in a modulation, the envelopes formed by the message signal begin to interfere with each other and become distorted. This can be tolerated in some cases, but with extreme distortion the signal may become unrecognizable. It was observed in the experiment that the change in frequency did not affect the amplitude modulation index. In demodulation, it has been observed that the signal begins to deteriorate when the modulation index exceeds 1. Distortion usually begins when the modulation index exceeds 1. Because at this point, the modulated signal rises above the amplitude of the carrier signal and the signal is crushed or cut off.

While we saw an impulse at the carrier frequency in DSB-LC amplitude modulation, we did not see it in DSB-SC. This showed us that the carrier was suppressed. LC AM demodulation has a simpler structure and is more suitable for multi-receiver systems, but it has disadvantages in terms of spectral efficiency. SC AM has advantages in terms of spectral efficiency and power efficiency, but its demodulation is more difficult. By performing modulation and demodulation processes in this experiment, the differences between LC and SC amplitude modulations were also observed.

## 5. References

**[1]** IIT JEE Study Materials, Amplitude Modulation, Accessed: 24.03.2024, https://byjus.com/jee/amplitude-modulation/

**[2]** Medium, Amplitude Modulation (AM) simulation with Python and NumPy, (16.09.2022), Accessed: 24.03.2024, https://mario-rodriguez.medium.com/amplitude-modulation-am-python-simulations-with-numpy-1fc50a4762af

**[3]** Tutorialspoint, Amplitude Modulation, Accessed: 24.03.2024, https://www.tutorialspoint.com/analog_communication/analog_communication_amplitude_modulation.htm

**[4]** Jim Lesurf, The Envelope Detector, Accessed: 24.03.2024, https://www.winlab.rutgers.edu/~crose/322_html/envelope_detector.html

**[5]** Wikipedia, Double-sideband suppressed-carrier transmission, Accessed: 24.03.2024, https://www.winlab.rutgers.edu/~crose/322_html/envelope_detector.html